TRANSPORTATION ENGINEERING · FINAL PROJECT

Mode Choice and Policy Welfare in Jabodetabek

An MNL → Nested Logit → Mixed Logit Comparison

Dhaneswaramandrasa
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Why Jabodetabek mode choice matters

SCALE

30M+ population

3.5 million daily commuters. Car/motorcycle-dominated mode split despite massive transit investment.

COVERAGE GAP

Spatial inequality

Transit expansion is spatially uneven: J1b (Parung) and J3b (Gading Serpong) have no formal transit at all.

WELFARE

Policy question

Which policies maximize welfare? How are gains distributed across zones and income segments?

Framework: 6-mode choice set → MNL / NL / MXL → AIC / LR / Wald selection → logsum ΔCS → 8 policy scenarios

Seven zones, six modes, synthetic population

Zones (7)	J1a–J5, Bodetabek → Jakarta CBD
Modes (6)	Car, Motorcycle, KRL, TransJakarta, RoyalTrans, MRT
Nest structure	Transit {KRL, MRT, TJ, Royal} · Private {Car, Moto}
Population	5,000 synthetic persons; 3 income segments (BPS Susenas 2023)
Transit LOS	r5py GTFS routing (AM peak 07:00–09:00, fallback 180 min)
Private LOS	BPR free-flow speeds (toll 80, arterial 40, local 25 km/h)
DGP	NL GEV closed-form choice generation, μ=25 scale normalization

β calibration: $\beta_{time,m}$ derived from Ilahi et al. (2021) Table 11 mode-specific VTTS via $\beta_{time} = \beta_{cost} \cdot VTTS / 60{,}000$. $\beta_{cost} = -1.42$. VOT preserved exactly under μ=25 scale normalization (Train 2009 §3.7).

MNL: specification & IIA problem

$$V_{mn} = ASC_m + \beta\_time,m \cdot T_{kn} + \beta\_cost \cdot C_{kn}$$

12 parameters · L-BFGS-B MLE · KRL = reference alternative

Parameter recovery	12/12 within 2 SE
ρ^2	0.280
AIC	10,121.65

IIA IS VIOLATED

Red-bus/blue-bus: clone KRL as "KRL Express" → transit share nearly doubles (11.7% → 21.8%)

NL cross-elasticity: within-nest substitution 1.67× cross-nest. MNL forces equal proportional substitution.

MNL fails to capture the correlation among transit alternatives → need Nested Logit.

NL: $\hat{\lambda} = 0.763$ — nesting confirmed

$$P_n(m) = P_n(m|k) \cdot P_n(k), \qquad P_n(k) = \exp(\lambda I_{kn}) / \Sigma_l \exp(\lambda I_{ln})$$

Two nests · 13 parameters · FIML via L-BFGS-B

$\hat{\lambda}$	0.763 ± 0.068
95% CI	[0.627, 0.900] — excludes 1.0
LR test vs MNL	$\chi^2 = 8.57$, df = 1, p = 0.003
Recovery	13/13 within 2 SE
AIC / BIC	10,115.08 / 10,199.80

KEY FINDING

$\lambda = 1$ rejected at $p < 0.01$

Transit modes share unobserved attributes (crowding, schedule coordination, station environment) that make them closer substitutes than IIA allows.

MXL: $\hat{\sigma}$ = 0.010 — no taste heterogeneity

Random β_cost ~ N(μ, σ) · R = 100 Halton draws (base=2)

H_0: σ_cost = 0 — any heterogeneity beyond NL nest correlation?

Parameter	Estimate	SE
σ_cost	0.010	0.033
μ_cost	−0.037	0.183

Wald: W = 0.091, p = 0.763 → fail to reject H_0

Positive control: Mixed-DGP (σ_true=0.02) → Wald p ≈ 0 ✓

Finding: No taste heterogeneity signal. MXL LL (−5,048.79) ≈ MNL LL (−5,048.83). Adding random β_cost gains 0.03 LL units — negligible.

NL captures the departure from IIA that matters.

NL wins on every criterion

Criterion	MNL	NL	MXL
K	12	13	13
LL($\hat{\beta}$)	−5,048.83	−5,044.54	−5,048.79
AIC	10,121.65	10,115.08	10,123.59
BIC	10,199.86	10,199.80	10,208.31
LR vs MNL (p)	—	0.003	0.799
Wald σ=0 (p)	—	—	0.763

NL selected: ΔAIC = −6.6 vs MNL, −8.5 vs MXL. LR rejects IIA at $p<0.01$. Wald fails to reject σ=0. Parsimonious correct specification → best_model.json

Logsum consumer surplus from NL

$EMU_n = \ln \Sigma_k \exp(\lambda \cdot I_{kn}), \quad I_{kn} = \ln \Sigma_{j \in k} \exp(V_{jn} / \lambda)$

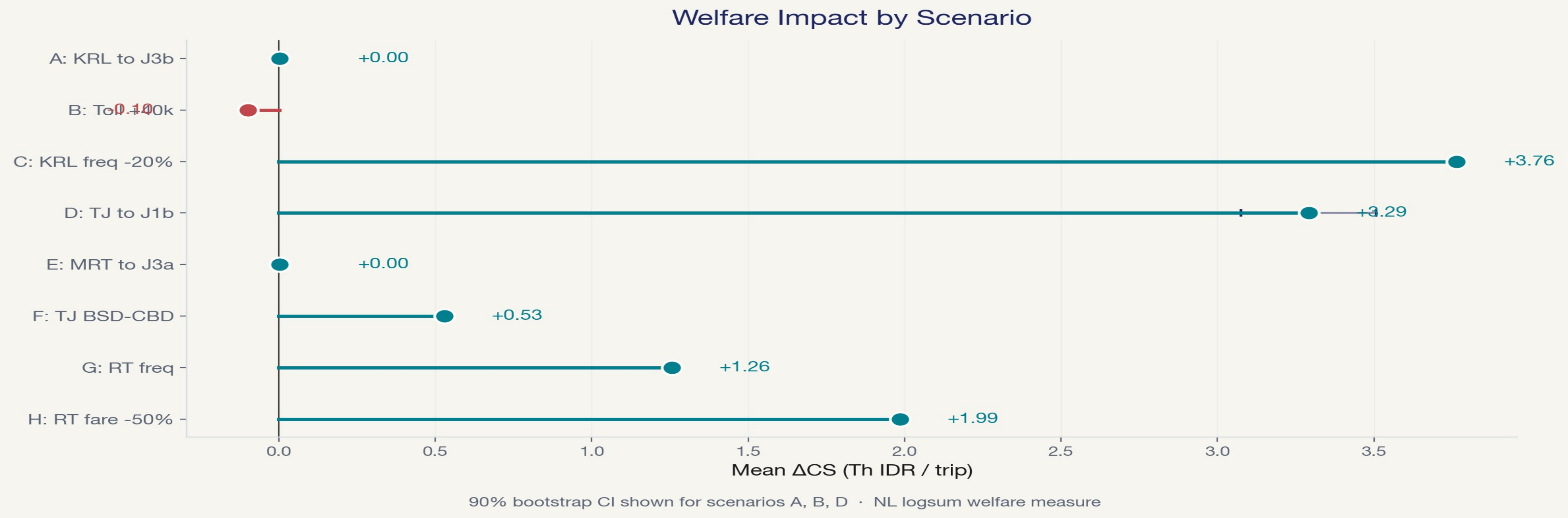
$CS_n = EMU_n / |\beta\_cost|, \quad \Delta CS_n = (EMU^{policy} - EMU^{baseline}) / |\beta\_cost|$

$\Delta W\_annual = \Sigma_n \Delta CS_n \times 250$

Baseline CS	P10	P90	β_cost (NL)
−53.65	−115.30	−9.48	−0.077
Th IDR/trip	Th IDR/trip	Th IDR/trip	SE = 0.097

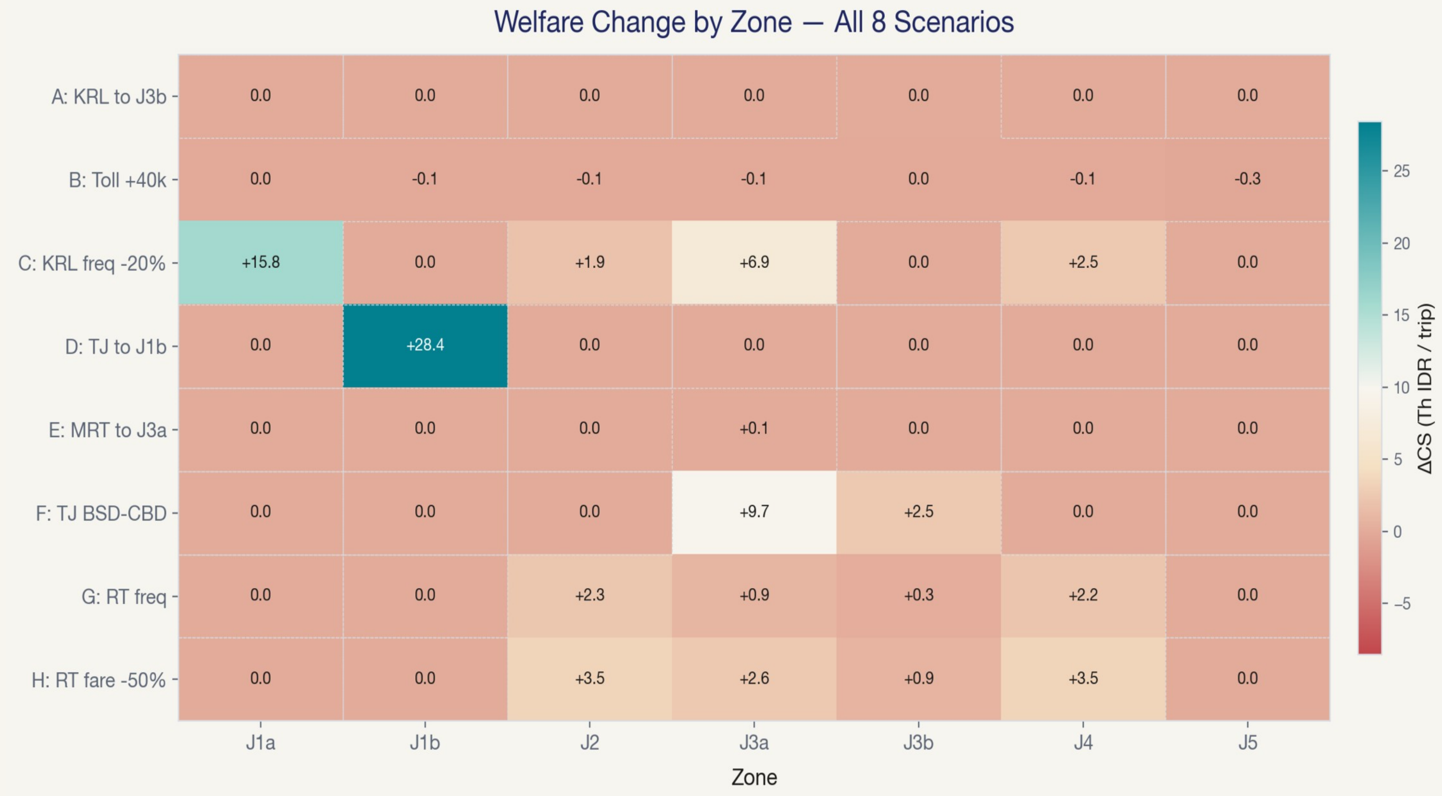
Bootstrap CIs use truncated-Normal draws (upper bound = −0.3·| $\hat{\beta}$|) to bound welfare estimates.

Eight scenarios — aggregate ΔCS comparison



Top 3 by ΔCS: C (KRL freq, +3.76) > D (TJ→J1b, +3.29) > H (RT fare, +1.99). **Regressive:** B (toll, −0.10, 0 winners)

Who wins and who is left out?



SC C

Maximizes aggregate welfare but J1b/J3b receive zero — no rail access.

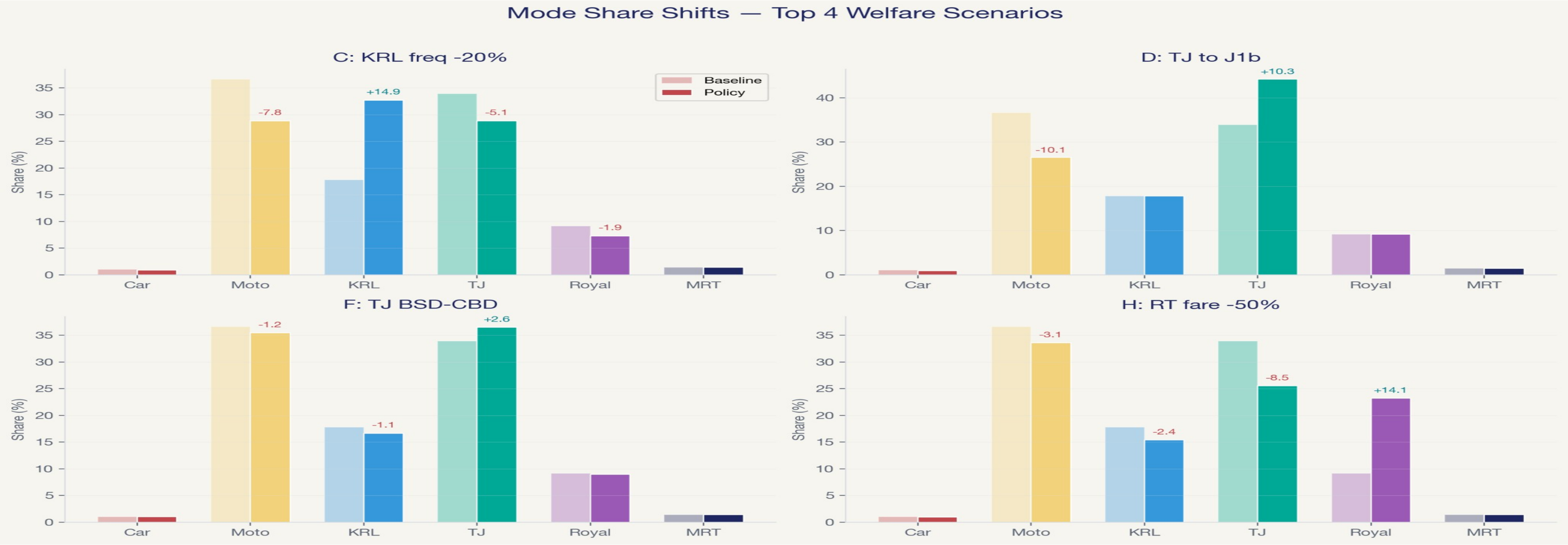
SC D

Concentrates gains in transit-orphan J1b (+28.43 Th IDR). TJ draws from motorcycles (−10.1 pp), not other transit.

SC F

Benefits both J3a (+9.66) and J3b (+2.49) through TJ BSD→CBD route restructuring.

Where do new riders come from?



Budget transit (TJ at Rp 3,500) draws primarily from motorcycles, not from other transit modes — expanding the pie rather than reshuffling it.

Four takeaways

1

NL is the appropriate model

$\hat{\lambda}$ = 0.763, LR $p = 0.003$, MXL adds no signal (Wald $p = 0.763$)

2

Top aggregate gain: Sc C

KRL frequency improvement: +3.76 Th IDR/trip, +6,580 Bn IDR/year

3

Most pro-equity: Sc D

TJ extension to J1b: +3.29 Th IDR/trip — concentrated in the most underserved zone

4

Bundle quality + expansion

Service quality improvements alone leave transit orphans behind. Combine with network expansion for equitable gains.

Thank You

Questions?

BACKUP A

Why scale parameter μ = 25?

Scale identification theorem (Train 2009 §3.7):

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True utility: U = V + ε, where ε ~ Gumbel(0, μ)

Normalized: Ũ = Ṽ + ε̃, where ε̃ ~ Gumbel(0, 1)

All coefficients compressed: β̃ = β / μ

VOT = β_time / β_cost is preserved — μ cancels
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Without μ = 25

Utility gaps too large → choice becomes deterministic (99%+ choosing one mode). No meaningful choice model.

With μ = 25

β estimates have larger SEs (the tradeoff), but Gumbel noise produces realistic choice distributions: Moto 36.7%, TJ 34.0%, KRL 17.8%.

Why drop ride-hailing & LRT?

2WRH (GoRide/GrabBike)

Ilahi $\beta\_time\_2wrh = -5.10$ estimated on short urban trips → near-zero share on 30–105 km commutes. Parameter non-transferability (Wardman et al. 2016).

4WRH (GoCar/GrabCar)

ASC calibration required >+40 utility units to match BPS aggregate share — outside defensible range.

LRT Jabodebek

Available in only 1 of 7 zones (J2). Single-zone presence → ASC absorbs zone unobservables, not mode preference.

Nine-mode DGP tested → six-mode adopted. Documented in report §2.2, §6.3.